

DELHI PUBLIC SCHOOL
SUBJECT- MATH CLASS-8

- Q1.** Define a "perfect square" and explain how it is different from a regular number.
- Q2.** Without calculating the actual square, explain how you can immediately tell that the number 2,048 is not a perfect square.
- Q3.** What is a "perfect cube"? Provide one example and explain why it fits this definition.
- Q4.** Explain the relationship between the number of zeros at the end of a natural number and the number of zeros in its square.
- Q5.** Briefly describe what the Sanskrit term *varga-mula* represents in modern mathematics and what it literally translates to.
- Q6.** A square-shaped community garden in Mumbai has an area of **144 square meters**. What is the length of one side of the garden?
- Q7.** A professional audio equipment crate is a perfect cube. If the length of one side is **4 feet**, what is the total volume of the crate in cubic feet?
- Q8.** You are tiling a square studio floor that is **15 feet** long. How many square-foot tiles do you need to cover the entire floor?
- Q9.** A cubic water tank holds **343 cubic meters** of water when full. What is the depth (height) of the tank?
- Q10.** A square photo frame has an area of **400 square centimeters**. You want to put a decorative ribbon around the entire edge. How much ribbon do you need?
- Q11.** Which is larger: $\sqrt{225}$ or $\sqrt[3]{1000}$?
- Q12.** If you double the side length of a square, how many times larger does the **area** become?
- Q13.** A cube of side 4 cm is made of tiny 1 cm unit cubes. Explain how many unit cubes are in each layer and how this leads to the total "cubic" value.
- Q14.** A square has an area of 81 cm^2 . If you cube the length of one of its sides, what is the resulting value?
- Q15.** A number is squared, and then **10** is added to the result to get **131**. What was the original number?